Date	03 July 2026
Team ID	
Team Size	3
Team Leader	Chekali Basavaraju
Team Member 1	Girish C
Team Member 2	Shaik Sameer
Team Member 3	
Team Member 4	
Project Name	Smart Lender: Machine Learning-Based Loan Approval Prediction System
Maximum Marks	

Problem—Solution Fit

Introduction

Problem	Impact	Pmposed Solution	Expected Benefit
Manual loan application screening is time-consuming.	Application processing is delayed, increasing operational workload.	Automate application screening using Machine Learning classification models.	Faster appmval decisions with leduced manual effort.
Human decision-making may introduce inconsistency and bias.	Applicants with similar financial profiles may meceive different	Use trained Machine l_eaming models to ensure consistent prediction based on historical	Improved fairness, consistency, and decision accuracy.
Large volumes Of applications are difficult to process	Reduced productivity and increased processing time	Develop a scalable web-based prediction platform integrated with an optimized Machine Leaning model.	High-speed processing with improved operational efficiency.
Applicants do not know their eligibility before submitting applications.	High rejection rates and poor customer experience	Provide an online eligibility prediction system before formal submission.	Better customer awareness and fewer unnecessary applications.

Sensitive applicant information requires secure handling.	Risk Of data breaches and privacy violations.	Implement authentication, mlebased access control, encrypted communication, and secure database storage.	Enhanced security, confidentiality, and regulatoty compliance.

The Problem—Solution Fit explains how the proposed A-'lachine Lærning—Based Cledit Cald Approval Pl?diction System dilectly addresses the challenges faced by financial institutions during the credit card approval process. It demonstrates how the proposed solution satisfies stakeholder needs by providing faster, more accurate, secure, and intelligent approval predictions.

Problem–Solution Fit Summary ⍰

Automates loan approval prediction

- – Improves prediction accuracy using Machine Learning
- – Reduces application processing time
- – Enhances customer experience
- – Ensures secure handling of applicant data
- – Supports scalable deployment using Flask and cloud technologies
- – Assists financial institutions in making reliable and data-driven decisions

Conclusion

The proposed Smart Lender Loan Approval Prediction System effectively overcomes the limitations of traditional manual loan evaluation by integrating intelligent Machine Learning algorithms, secure web technologies, and automated workflows. The system provides faster, more consistent, and reliable loan approval decisions while improving operational efficiency and overall customer satisfaction..